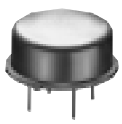
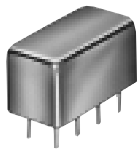
LOW-NOISE AMPLIFIERS 50Ω

BROADBAND, LINEAR 0.1 to 3000 MHz

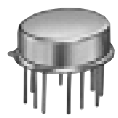


AMP

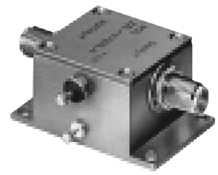


MAN

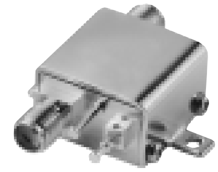
ZFL



TO



ZEL



ZX60

up to +16 dBm output

MODEL NO.	FREQ. (MHz) f_L-f_U	NF (dB) Typ.	GAIN (dB) Min.	GAIN Flatness Max. m	GAIN Flatness Max. Total range	MAXIMUM POWER (dBm) Output (1 dB Comp.)	MAXIMUM POWER (dBm) Input (no damage)	INTERCEPT POINT (dBm) IP3 Typ.	VSWR Typ. In	VSWR Typ. Out	DC POWER Volt (V)	DC POWER Current (mA)	CASE STYLE Note B	CONNECTION	PRICE $ Qty. (1-9)
AMP-15	5-1000	2.8	13	±0.6	±1.2	+8	+13	+22	2:1	2:1	15	29	PP120	cd	49.95
AMP-75	5-500	2.4	19	±0.4	±1.0	+12	+13	+28	2:1	2:1	15	31	PP120	cd	49.95
AMP-76	5-500	3.1	26	±0.7	±1.0	+13.5	+6	+28	2:1	2:1	15	71	PP120	cd	78.95
AMP-77	5-500	3.3	15	±0.4	±1.0	+16	+13	+32	2:1	2:1	15	56	PP120	cd	55.95
MAN-1LN**	0.5-500	3.0	28	±0.5	±1.4	+7	+15	+18	1.8:1	1.8:1	12	60	A05	cc	19.95
MAN-1HLN	10-500	3.7	10	±0.5	±0.8	+15	+15	+30	1.8:1	1.8:1	12	70	A06	cc	19.95
ZFL-500HLN	10-500	3.8	19	—	±0.4	+16	+15	+30	2:1	2:1	15	110	Y460	—	99.95
ZFL-500LN*	0.1-500	2.9	24	—	±0.5	+5	+5	+14	1.5:1	1.6:1	15	60	Y460	—	79.95
ZFL-1000LN	0.1-1000	2.9	20	—	±0.5	+3	+5	+14	1.5:1	2:1	15	60	Y460	—	89.95
NEW ZX60-3011	400-1000	1.4	12	—	±.70	+19.5	+15	+31	1.7:1	1.6:1	12	120	GC957	—	139.95
	1000-1700	1.5	11	—	±1.0	+19.5	+15	+31	1.7:1	1.6:1	12	120			
	1700-2400	1.7	9	—	±1.0	+18.5	+15	+31	1.7:1	1.6:1	12	120			
	2400-3000	1.8	7.5	—	±.70	+18.0	+15	+31	1.7:1	1.6:1	12	120			

m = mid range [2 f_L to f_U/2]

features

- very low noise
- ideal for printed-circuit designs (AMP,MAN & TO series)
- high dynamic range (ZHL- HLN, ZQL & ZX60 series)
- smooth response over entire band, no external resonances
- low impedance, less susceptible to EMI
- easy to use, 50 ohm input/output
- all models are cascadable

NOTES:

* VSWR 1.6:1 maximum from 0.1 to 0.2 MHz. Also available with BNC connectors.

** Below 5 MHz, 1 dB compression point decreases to 6.5 dBm.

▲ Available only with SMA connectors

B. Connector types and case mounted options, case finishes are given in section 0, see "Case styles & outline drawings".

C. Prices and specifications subject to change without notice.

D. For Quality Control Procedures see Table of Contents, Section 0, "Mini-Circuits Guarantees Quality" article. For Environmental Specifications see Amplifier Selection Guide.

1. Absolute maximum power, voltage and current rating:
   1a. AMP models, 17V DC. 1b. MAN models, 12.5V DC.
   1c. ZQL models, 17V DC. 1d. ZX60, 15VDC 1e. ZQLSC, 36VDC
2. Open load is not recommended, potentially can cause damage. With no load, derate max input power by 20 dB.
3. ZEL and TO models, NF specified at room temperature, increases to 2 dB typical at +85 deg.C. (TO-0812LN increases to 1.6 dB max.)
4. ZHL models, NF specified at room temperature, increases to 2.3 dB maximum at +65 deg.C.

NSN GUIDE

MCL NO.	NSN
AMP-15	5996-01-350-9550
AMP-75	5996-01-350-9551
AMP-77	5996-01-350-9549
ZEL-1724LN	5996-01-450-0781
ZFL-1000LN	5996-01-412-3031
ZHL-0812HLN	5996-01-453-2464

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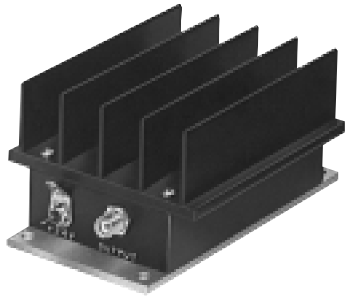
ZQLSC

ZQL



ZHL-case S32

ZHL-case NN92

up to +27 dBm output

MODEL NO.	FREQ. (MHz) f_L-f_U	NF (dB) Max.	GAIN (dB) Min.	GAIN (dB) Flatness Max.	MAXIMUM POWER (dBm) Output (1 dB Comp.) Typ.	MAXIMUM POWER (dBm) Input (no damage)	INTERCEPT POINT (dBm) IP3 Typ.	VSWR Max. In	VSWR Max. Out	DC POWER Volt (V)	DC POWER Current (mA)	CASE STYLE Note B	CONNECTION	PRICE $ Qty. (1-9)
TO-0812LN	800-1200	1.2	20	±1.0	+8	+10	+22.5	2.5:1	2.5:1	15	70	QQ96	ce	199.00
TO-1217LN	1200-1700	1.6	20	±1.0	+10	+13	+25	2.5:1	2.5:1	15	70	QQ96	ce	199.00
TO-1724LN	1700-2400	1.6	20	±1.0	+10	+13	+22	2.5:1	2.5:1	15	70	QQ96	ce	199.00
▲ ZEL-0812LN	800-1200	1.5	20	±1.0	+8	+13	+18	2.5:1	2.5:1	15	70	EEE132	—	274.95
▲ ZEL-1217LN	1200-1700	1.5	20	±1.0	+10	+13	+25	2.5:1	2.5:1	15	70	EEE132	—	274.95
▲ ZEL-1724LN	1700-2400	1.5	20	±1.0	+10	+13	+22	2.5:1	2.5:1	15	70	EEE132	—	274.95
▲ ZHL-0812MLN	800-1200	1.6	28	±1.0	+20	0	+33	2.5:1	2.5:1	15	300	S32	—	295.00
▲ ZHL-1217MLN	1200-1700	1.5	30	±1.0	+20	0	+34	2.5:1	2.5:1	15	300	S32	—	295.00
▲ ZHL-1724MLN	1700-2400	1.5	28	±1.0	+20	0	+32	2.5:1	2.5:1	15	300	S32	—	295.00
▲ ZHL-0812HLN	800-1200	1.5	30	±1.0	+26	+10	+36	2.4:1	2.4:1	15	725	NN92	—	399.50
▲ ZHL-1217HLN	1200-1700	1.5	30	±1.0	+26	+10	+36	2.4:1	2.4:1	15	725	NN92	—	399.50
▲ ZHL-1724HLN	1700-2400	1.5	30	±1.0	+26	+10	+36	2.4:1	2.4:1	15	725	NN92	—	399.50
NEW ZQLSC-1100	600-824	1.1	17.5	±1.0	+18.5	+10	+32.5	2.0:1	2.0:1	24	185	GZ1067	pt	295.00
	824-849	0.7	17.5	±0.3	+19	+10	+34	1.7:1	1.7:1	24	185			
	850-915	0.8	17.0	±0.4	+19	+10	+35	1.8:1	1.8:1	24	185			
	915-1100	1.0	15.0	±1.2	+19	+10	+35.5	1.8:1	1.8:1	24	185			
NEW ZQLSC-2400	1400-1850	1.15	13.5	±2.5	+19	+10	+37	1.7:1	1.6:1	24	185	GZ1067	pt	295.00
	1850-2000	1.25	12.5	±1.0	+20.5	+10	+35	1.5:1	1.6:1	24	185			
	2000-2400	1.65	10.5	±2.0	+20	+10	+34	1.6:1	1.9:1	24	185			
								Typ.						
ZQL-900LNW	800-900	1.6	13	±1.6	+21	+10	+35	1.2:1	1.1:1	15	160	CW686	—	229.00
ZQL-900LN	824-849	1.3	15	±0.5	+21	+10	+35	1.2:1	1.1:1	15	160	CW686	—	229.00
ZQL-1900LNW	1700-2000	1.6	14	±1.8	+18.5	+10	+37	1.15:1	1.25:1	15	160	CW686	—	249.00
ZQL-1900LN	1850-1910	1.5	15	±0.5	+19	+10	+37	1.15:1	1.25:1	15	160	CW686	—	249.00
ZQL-900MLNW	800-900	1.7	22	±2.2	+23	+3	+41	1.3:1	1.4:1	15	230	CW686	—	249.00
ZQL-900MLN	824-849	1.5	25.5	±0.5	+24.5	+3	+41	1.3:1	1.4:1	15	230	CW686	—	249.00
ZQL-1900MLNW	1800-2000	1.6	23	±2.0	+25	+3	+41	1.4:1	1.25:1	15	310	CW686	—	265.00
ZQL-1900MLN	1850-1910	1.5	25	±0.7	+26	+3	+41	1.25:1	1.2:1	15	310	CW686	—	265.00
ZQL-2700MLNW	2200-2400	1.3	25	±1.0	+25	+3	+38	1.25:1	1.15:1	15	325	CW686	—	281.95
	2200-2700	1.5	25	±2.3	+25	+3	+38	1.25:1	1.15:1	15	325			

pin connections

PORT	cc	cd	ce	pt
RF IN	1	2	5	J1
RF OUT	8	4	11	J2
DC	5	1	2	5
TTL ALARM OUTPUT	—	—	—	1
GND TO TEST ALARM, normally open	—	—	—	7*,9*
CASE GND	2,3,4,6	3	1,3,4,6,7,8,9,10,12	2,4
NOT USED	7	—	—	3,6,8

*Grounding Pin 7 will sink 75mA of current through Pin 7 creating a high-current alarm condition inside the amplifiers. A red LED and TTL high output will occur. Pin 7 floats at +4.3V typ. when open.

*Grounding Pin 9 will sink 2mA of current through pin 9 and creating a low-current alarm condition inside the amplifier. A red LED and TTL high output will occur. Pin 9 floats at about +0.6V typ. when open.

Alarm Functions for ZQLSC Series

Normal:	TTL low output (0 to 0.8V), green LED
Alarm:	TTL high output (4 to 5V), red LED
DC & alarm connector:	9-pin male D-sub

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